

Computing the interleaving distance between two interval modules

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I found nowhere the explicit computation of the interleaving distance between interval modules, though it is well known that it equals the bottleneck distance.

1 Preliminaries

Let $\mathbf{k}$ be an arbitrary field, fixed throughout. We denote by $\mathbb{R}$ either the real line or the category induced by the poset $(\mathbb{R}, \leq)$. All functors are covariant.

1.1 The category of persistence modules

A **persistence module** is a functor $\mathbb{R} \rightarrow \text{Vect}(\mathbf{k})$.

Example 1 (Important). Let $I \subseteq \mathbb{R}$ be an interval. Define $\mathcal{I}: \mathbb{R} \rightarrow \text{Vect}(\mathbf{k})$ by

$$t \mapsto \begin{cases} \mathbf{k} & \text{if } t \in I \\ 0 & \text{else} \end{cases} \quad \text{and} \quad r \rightarrow s \mapsto \begin{cases} \mathbf{k} \xrightarrow{\text{Id}} \mathbf{k} & \text{if } r, s \in I \\ 0 & \text{else} \end{cases}$$

on objects and morphisms, respectively. We call $\mathcal{I}$ the **interval module** over I .

Remark. Interval modules are the building blocks of persistence modules. In fact, one objective of persistence theory is to understand persistence modules by decomposing them into interval modules. However, this is not always possible. A situation when such a decomposition is guaranteed is when we work with finite dimensional vector spaces, that is, if we work over $\text{vect}(\mathbf{k})$ instead of $\text{Vect}(\mathbf{k})$.

The concise definition of persistence module can be expanded to aide comprehension. Let $V: \mathbb{R} \rightarrow \text{Vect}(\mathbf{k})$ be a persistence module.

- For each $t \in \mathbb{R}$ and each arrow $r \rightarrow s$ denote $V(t) = V_t$ and $V(r \rightarrow s) = v_s^r$.
- Since V is a functor, we have the so-called **composition law**:

$$v_t^s \circ v_s^r = v_t^r \quad \text{whenever } r \leq s \leq t$$

$$\text{and } V(t \rightarrow t) = v_t^t = \text{Id}_{V_t}.$$

Thus, a persistence module V is a family of vector spaces $(V_t)_{t \in \mathbb{R}}$ together with a family $(v_t^s: V_s \rightarrow V_t \mid s \leq t)_{s,t \in \mathbb{R}}$ of $\mathbf{k}$ -linear maps that satisfy the composition law.

We got a set theoretical definition from the categorical one above. In fact, the two are equivalent. We will use the first, but prefer the notation of the second.

Since persistence modules are functors, we can consider the natural transformations between them, and since those can be composed (using vertical composition) we obtain a category.

Definition 1. The **category of persistence modules** is $\mathbf{X} = \text{Fun}(\mathbb{R}, \text{Vect}(\mathbf{k}))$.

Yes, this category is fun. Concretely, a morphism of persistence modules $\alpha: U \rightarrow V$ is a family of linear maps $(\alpha_t: U_t \rightarrow V_t)_{t \in \mathbb{R}}$ such that the diagram

$$\begin{array}{ccc} U_s & \xrightarrow{u_t^s} & U_t \\ \alpha_s \downarrow & & \downarrow \alpha_t \\ V_s & \xrightarrow{v_t^s} & V_t \end{array}$$

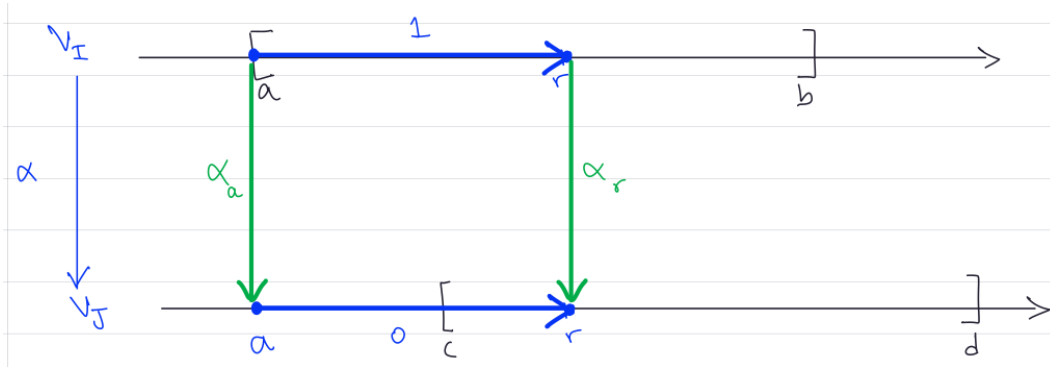
commutes for all $s \leq t$.

1.2 The shift functor

Let U be a persistence module. Define a new persistence module $U[t]$ by

$$U[t](r) = U_{t+r} \quad \text{and} \quad U[t](r \rightarrow s) = u_{t+r}^{t+s}.$$

We say that $U[t]$ is a **shifted module**. Suppose we've made two shifts of U , say $U[s]$ and $U[t]$, where $s \leq t$. We might ask ourselves: what is the relationship between these shifted modules? Since these are functors, we wonder if there is a canonical natural transformation $\alpha: U[s] \rightarrow U[t]$, and in fact there is.



The functor $T = [\cdot]: \mathbb{R} \rightarrow \text{End}(X)$ is defined as follows

- Objects: for each $t \in \mathbb{R}$ we have a functor $T(t): X \rightarrow X$ denoted T_t , which is defined by $T_t(U) = U[t]$ on objects $U \in X$, and by $T_t(\alpha) = (\alpha_{r+t})_{r \in \mathbb{R}}$ on morphisms $\alpha = (\alpha_r)_{r \in \mathbb{R}}$. (Equivalently, we can define $T_t(\alpha) = \alpha \star [t]$)
- Morphisms: for each $r \rightarrow s$, define $T(r \rightarrow s): T_r \rightarrow T_s$ as

1.3 The interleaving distance between persistence modules

Let U and V be persistence modules. A t -**interleaving** of U and V is a pair of morphisms (ϕ, ψ) that make the diagrams

$$\begin{array}{ccc}
 U & \xrightarrow{\mathbb{1}_U^{2t}} & U[2t] \\
 \searrow \phi & & \nearrow \psi[t] \\
 & V[t] &
 \end{array}
 \qquad
 \begin{array}{ccc}
 V & \xrightarrow{\mathbb{1}_V^{2t}} & V[2t] \\
 \searrow \psi & & \nearrow \phi[t] \\
 & U[t] &
 \end{array}$$

commute. If such a pair exists, we say that U and V are t -interleaved.

Definition 2. The interleaving distance between two intervalmodules is

$$d_T(U, V) = \inf \{ t \geq 0 \mid \text{there exists a } t\text{-interleaving of } U \text{ and } V \}.$$

2 The computation